

Solutions to Diffusion Models and Uncertainty Problems

ELG 5218 – Uncertainty Evaluation in Engineering Measurements and Machine Learning

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PART A: SOLUTIONS

A1. Score Function Interpretation

- (a) $s(\mathbf{x}) = \nabla_{\mathbf{x}} \log p(\mathbf{x})$ is a vector pointing in the direction of steepest increase of the log-density at $\mathbf{x}$. Geometrically, it points toward nearby regions of higher probability. Its magnitude indicates how steeply the density rises.
- (b) $\log p(\mathbf{x}) = \log \tilde{p}(\mathbf{x}) - \log Z$, where Z is the normalizing constant. Since Z is independent of $\mathbf{x}$, $\nabla_{\mathbf{x}} \log p(\mathbf{x}) = \nabla_{\mathbf{x}} \log \tilde{p}(\mathbf{x})$. The gradient eliminates the intractable partition function.

A2. Noise Prediction vs. x_0 -Prediction vs. Score

- (a) From the forward process $\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon}$, we can solve for any quantity:

$$\begin{aligned}\boldsymbol{\epsilon} &= \frac{\mathbf{x}_t - \sqrt{\bar{\alpha}_t} \mathbf{x}_0}{\sqrt{1 - \bar{\alpha}_t}}, \\ \mathbf{x}_0 &= \frac{\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon}}{\sqrt{\bar{\alpha}_t}}, \\ \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t | \mathbf{x}_0) &= -\frac{\mathbf{x}_t - \sqrt{\bar{\alpha}_t} \mathbf{x}_0}{1 - \bar{\alpha}_t} = -\frac{\boldsymbol{\epsilon}}{\sqrt{1 - \bar{\alpha}_t}}.\end{aligned}$$

Hence predicting any one determines the other two via linear algebra.

A3. Diffusion vs. Kalman Filtering

- (a) **Similarities:** (1) Both involve sequential Gaussian updates: the Kalman filter updates $p(\mathbf{x}_t | \mathbf{y}_{1:t})$ step by step; DDPM updates $p(\mathbf{x}_{t-1} | \mathbf{x}_t)$. (2) Both maintain a mean and covariance (or implicit equivalent via the neural network) at each step.
- (b) **Differences:** (1) The Kalman filter uses a *known linear* dynamical model and measurement model; DDPM uses a *learned nonlinear* denoiser. (2) The Kalman filter runs

forward in time incorporating new observations; DDPM runs *backward* from noise to data. (3) The Kalman filter is exact for linear-Gaussian systems; DDPM approximates arbitrary distributions.

PART B: SOLUTIONS

B1. Closed-Form Forward Process

- (a) **Base case** ($t = 1$): $\mathbf{x}_1 = \sqrt{\alpha_1} \mathbf{x}_0 + \sqrt{1 - \alpha_1} \boldsymbol{\epsilon}_1$, so $q(\mathbf{x}_1 | \mathbf{x}_0) = \mathcal{N}(\sqrt{\alpha_1} \mathbf{x}_0, (1 - \alpha_1)\mathbf{I})$ and $\bar{\alpha}_1 = \alpha_1$. Correct.

Inductive step. Assume $q(\mathbf{x}_{t-1} | \mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_{t-1}} \mathbf{x}_0, (1 - \bar{\alpha}_{t-1})\mathbf{I})$. Then

$$\begin{aligned} \mathbf{x}_t &= \sqrt{\alpha_t} \mathbf{x}_{t-1} + \sqrt{1 - \alpha_t} \boldsymbol{\epsilon}_t \\ &= \sqrt{\alpha_t} (\sqrt{\bar{\alpha}_{t-1}} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_{t-1}} \boldsymbol{\epsilon}'_{t-1}) + \sqrt{1 - \alpha_t} \boldsymbol{\epsilon}_t. \end{aligned}$$

The mean coefficient is $\sqrt{\alpha_t \bar{\alpha}_{t-1}} = \sqrt{\bar{\alpha}_t}$. The variance is $\alpha_t(1 - \bar{\alpha}_{t-1}) + (1 - \alpha_t) = \alpha_t - \alpha_t \bar{\alpha}_{t-1} + 1 - \alpha_t = 1 - \bar{\alpha}_t$. Since both noise terms are independent Gaussians, their sum is Gaussian: $q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t)\mathbf{I})$. $\square$

- (b) $(\sqrt{\bar{\alpha}_t})^2 + (1 - \bar{\alpha}_t) = \bar{\alpha}_t + 1 - \bar{\alpha}_t = 1$. This variance-preserving property ensures $\mathbf{x}_t$ remains unit-variance when $\mathbf{x}_0$ is normalized, keeping the network's input scale controlled across all timesteps.
- (c) As $T \rightarrow \infty$, $\bar{\alpha}_T \rightarrow 0$ (since each $\alpha_s < 1$), so $q(\mathbf{x}_T | \mathbf{x}_0) \rightarrow \mathcal{N}(\mathbf{0}, \mathbf{I})$. The data structure is completely destroyed and the distribution becomes isotropic Gaussian, which is the starting point for reverse sampling.

B2. Reverse Process Mean as a Denoiser

- (a) B2(a) Oracle reverse mean as a denoising step. We start from the deterministic part of the reverse update

$$x_{t-1}^{\text{det}} = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_{\theta}(x_t, t) \right).$$

If the predictor is perfect, then

$$\epsilon_{\theta}(x_t, t) = \epsilon,$$

where the forward process satisfies

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon.$$

Substituting this expression for x_t into the reverse update gives

$$x_{t-1}^{\text{det}} = \frac{1}{\sqrt{\alpha_t}} \left(\sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon \right).$$

Now combine the two ϵ -terms:

$$x_{t-1}^{\text{det}} = \frac{1}{\sqrt{\alpha_t}} \left(\sqrt{\bar{\alpha}_t} x_0 + \left[\sqrt{1 - \bar{\alpha}_t} - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \right] \epsilon \right).$$

Putting the bracketed term over a common denominator,

$$\sqrt{1 - \bar{\alpha}_t} - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} = \frac{1 - \bar{\alpha}_t - (1 - \alpha_t)}{\sqrt{1 - \bar{\alpha}_t}} = \frac{\alpha_t - \bar{\alpha}_t}{\sqrt{1 - \bar{\alpha}_t}}.$$

Therefore,

$$x_{t-1}^{\text{det}} = \frac{\sqrt{\bar{\alpha}_t}}{\sqrt{\alpha_t}} x_0 + \frac{\alpha_t - \bar{\alpha}_t}{\sqrt{\alpha_t} \sqrt{1 - \bar{\alpha}_t}} \epsilon.$$

Using

$$\bar{\alpha}_t = \alpha_t \bar{\alpha}_{t-1},$$

we obtain

$$\frac{\sqrt{\bar{\alpha}_t}}{\sqrt{\alpha_t}} = \sqrt{\bar{\alpha}_{t-1}}.$$

Hence

$$x_{t-1}^{\text{det}} = \sqrt{\bar{\alpha}_{t-1}} x_0 + \frac{\alpha_t - \bar{\alpha}_t}{\sqrt{\alpha_t} \sqrt{1 - \bar{\alpha}_t}} \epsilon.$$

This shows that x_{t-1}^{det} is formed from the same clean sample x_0 and the same noise realization ϵ as x_t , but with a reduced noise contribution. Since

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon,$$

the deterministic reverse step moves the sample from the noisier point x_t toward the clean sample x_0 . In this sense, the reverse mean acts as a denoising step.

- (b) The noise coefficient in $\mathbf{x}_t$ is $\sqrt{1 - \bar{\alpha}_t}$; after one reverse step the effective noise level drops to $\sqrt{1 - \bar{\alpha}_{t-1}}$. The fraction of noise removed per step is $1 - \sqrt{(1 - \bar{\alpha}_{t-1})/(1 - \bar{\alpha}_t)}$, which is small for each individual step but accumulates over all T steps.
- (c) If $\sigma_t = 0$ for all t , the reverse process becomes deterministic: for a fixed initial noise x_T , the whole sampling trajectory is uniquely determined. This is closely related to deterministic DDIM-style sampling.

B3. Denoising Score Matching

- (a) $q(\tilde{\mathbf{x}} | \mathbf{x}) = \mathcal{N}(\mathbf{x}, \sigma^2 \mathbf{I})$, so

$$\nabla_{\tilde{\mathbf{x}}} \log q(\tilde{\mathbf{x}} | \mathbf{x}) = -\frac{\tilde{\mathbf{x}} - \mathbf{x}}{\sigma^2} = -\frac{\boldsymbol{\epsilon}}{\sigma}$$

since $\tilde{\mathbf{x}} = \mathbf{x} + \sigma \boldsymbol{\epsilon}$.

- (b) The DSM objective is

$$\mathbb{E}_{\mathbf{x}, \boldsymbol{\epsilon}} \left[\|s_{\theta}(\tilde{\mathbf{x}}) - \nabla_{\tilde{\mathbf{x}}} \log q(\tilde{\mathbf{x}} | \mathbf{x})\|^2 \right] = \mathbb{E}_{\mathbf{x}, \boldsymbol{\epsilon}} \left[\|s_{\theta}(\mathbf{x} + \sigma \boldsymbol{\epsilon}) + \frac{\boldsymbol{\epsilon}}{\sigma}\|^2 \right].$$

This is exactly the stated form, confirming equivalence.

- (c) At the optimum, $s_{\theta}(\tilde{\mathbf{x}})$ minimizes the expected squared error to $-\boldsymbol{\epsilon}/\sigma = \nabla_{\tilde{\mathbf{x}}} \log q(\tilde{\mathbf{x}} | \mathbf{x})$. By the Vincent/DSM theorem, this optimal s_{θ} also equals $\nabla_{\tilde{\mathbf{x}}} \log q_{\sigma}(\tilde{\mathbf{x}})$, i.e. the score of

the noise-convolved data distribution. Thus noise-prediction training implicitly learns the score field.

B4. From ELBO to the Simplified Loss

- (a) **First term** $D_{\text{KL}}(q(\mathbf{x}_T | \mathbf{x}_0) \| p(\mathbf{x}_T))$: measures how close the terminal forward distribution is to the prior $\mathcal{N}(\mathbf{0}, \mathbf{I})$. This is constant (no learnable parameters) under standard schedules.

Middle terms $D_{\text{KL}}(q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) \| p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t))$: these are the denoising matching terms. Each asks the learned reverse step at time t to match the tractable posterior $q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0)$.

Last term $-\log p_\theta(\mathbf{x}_0 | \mathbf{x}_1)$: a reconstruction term measuring how well the final denoising step recovers the clean data.

- (b) The posterior $q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0)$ is Gaussian with:

$$\begin{aligned}\tilde{\mu}_t &= \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}} \beta_t}{1 - \bar{\alpha}_t} \mathbf{x}_0, \\ \tilde{\beta}_t &= \frac{(1 - \bar{\alpha}_{t-1})\beta_t}{1 - \bar{\alpha}_t}.\end{aligned}$$

- (c) The KL between two Gaussians with the same (fixed) variance reduces to a scaled squared distance between means: $D_{\text{KL}} = \frac{1}{2\tilde{\beta}_t} \|\tilde{\mu}_t - \mu_\theta(\mathbf{x}_t, t)\|^2$. Since $\tilde{\mu}_t$ depends on $\mathbf{x}_0$ (and hence on ϵ via the forward process), expressing μ_θ in terms of ϵ_θ and dropping the timestep-dependent weighting gives the simplified loss $\mathcal{L}_{\text{simple}} = \mathbb{E}_{t, \mathbf{x}_0, \epsilon} [\|\epsilon - \epsilon_\theta(\mathbf{x}_t, t)\|^2]$.

B5. Langevin Dynamics: Stationary Distribution

- (a) The deterministic component $\eta \nabla_{\mathbf{x}} \log p(\mathbf{x}_k)$ is a gradient ascent step on the log-density, pushing toward modes. The stochastic component $\sqrt{2\eta} \mathbf{z}_k$ injects noise for exploration. Without noise, the iteration becomes gradient ascent, converging to a mode (the MAP estimate) rather than sampling from the full distribution.

- (b) The *continuous-time* Langevin diffusion

$$dX_t = \nabla_x \log p(X_t) dt + \sqrt{2} dW_t$$

has $p(x)$ as its stationary distribution under suitable regularity conditions. However, the practical update

$$x_{k+1} = x_k + \eta \nabla_x \log p(x_k) + \sqrt{2\eta} z_k, \quad z_k \sim \mathcal{N}(0, I),$$

is a *discretized* version of this diffusion, usually called the unadjusted Langevin algorithm (ULA). For a fixed nonzero step size η , this discrete chain generally does not have exactly $p(x)$ as its stationary distribution; instead, it only approximates the target distribution, with the approximation improving as η becomes sufficiently small.

(c) For $p(\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}, \sigma^2 \mathbf{I})$:

$$\log p(\mathbf{x}) = -\frac{\|\mathbf{x} - \boldsymbol{\mu}\|^2}{2\sigma^2} + \text{const},$$

$$\nabla_{\mathbf{x}} \log p(\mathbf{x}) = -\frac{\mathbf{x} - \boldsymbol{\mu}}{\sigma^2}.$$

So the drift is $\eta \nabla_{\mathbf{x}} \log p(\mathbf{x}_k) = -\frac{\eta}{\sigma^2}(\mathbf{x}_k - \boldsymbol{\mu})$, a linear function that pulls $\mathbf{x}_k$ toward $\boldsymbol{\mu}$. This is an Ornstein–Uhlenbeck process in the continuous-time limit.

PART C: SOLUTIONS

C1. Effect of the Number of Steps T

- (a) If T is too small, each forward step adds a large amount of noise, and the Gaussian assumption on the reverse step $q(\mathbf{x}_{t-1} | \mathbf{x}_t)$ becomes a poor approximation (the true reverse is only approximately Gaussian for small noise increments). The learned reverse model struggles to undo large corruptions in one step.
- (b) If T is very large, inference requires T sequential neural-network evaluations, which is computationally expensive (e.g. $T = 4000$ at 50 ms per evaluation ≈ 200 s per sample).
- (c) DDIM reinterprets the forward process as non-Markovian, allowing the reverse chain to skip timesteps while reusing the same ϵ_θ . DPM-Solver uses higher-order ODE solvers to take larger but more accurate steps. Neither requires retraining; they simply change the inference schedule.

C2. Latent Diffusion: Effect of Compression

- (a) Diffusion steps in $\mathbb{R}^d$ instead of $\mathbb{R}^D$ have cost $O(d)$ vs. $O(D)$ per step. For images, D might be $256^2 \times 3 \approx 200\text{K}$ while d might be $32^2 \times 4 = 4\text{K}$, a $50\times$ reduction.
- (b) The autoencoder may lose fine-grained details (e.g. texture, high-frequency edges). Generated samples inherit these compression artifacts, bounding the achievable quality by the autoencoder’s fidelity.
- (c) The diffusion model operates purely in latent space; it cannot generate latent codes that decode to details the autoencoder discards. It can partially compensate by learning a latent distribution that avoids poorly-decoded regions, but fundamental autoencoder limitations remain.

PART D: SOLUTIONS

D1. Time-Series Forecasting with Diffusion

- (a) **Training:** Given a training window $(\mathbf{x}_{\text{past}}, \mathbf{x}_{\text{future}})$, the future portion is noised: $\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_{\text{future}} + \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon}$. The denoiser $\epsilon_\theta(\mathbf{x}_t, t, \mathbf{x}_{\text{past}})$ receives the noised future concatenated with (or conditioned on) the clean past. The loss is $\mathcal{L} = \mathbb{E}[\|\boldsymbol{\epsilon} - \epsilon_\theta(\mathbf{x}_t, t, \mathbf{x}_{\text{past}})\|^2]$.
- (b) **Point forecast:** sample mean $\hat{\mathbf{x}}_{\text{future}} = \frac{1}{S} \sum_{s=1}^S \mathbf{x}_{\text{future}}^{(s)}$ (or sample median for robustness).
90% prediction interval: For each time index j , sort the S samples and take the 5th and 95th percentiles as interval endpoints.
- (c) At horizon $t+1$, the observed past tightly constrains plausible values, so samples cluster. At horizon $t+24$, accumulated uncertainty (weather changes, demand shifts) allows a wider range of plausible futures. This is the natural growth of predictive uncertainty with horizon, analogous to the Kalman-filter covariance growing during prediction steps.
- (d) **Kalman filter:** assumes linear dynamics and Gaussian noise; exact and fast for such models but cannot represent multimodal futures. **Diffusion:** no linearity or Gaussianity assumption; can capture multimodal, non-Gaussian forecast distributions; but requires $T \times S$ neural-network evaluations, so it is much more expensive.

D2. Signal Reconstruction from Noisy Measurements

- (a) The diffusion model encodes a learned prior $p_\theta(\mathbf{x}_0)$ over plausible signals. Combined with the known likelihood $p(\mathbf{y} \mid \mathbf{x}_0) = \mathcal{N}(\mathbf{A}\mathbf{x}_0, \sigma_n^2 \mathbf{I})$, Bayes' rule gives $p(\mathbf{x}_0 \mid \mathbf{y}) \propto p(\mathbf{y} \mid \mathbf{x}_0) p_\theta(\mathbf{x}_0)$. The diffusion model replaces a hand-crafted prior (e.g. sparsity) with a data-driven one.
- (b) At each reverse step t , after computing the unconditional update $\hat{\mathbf{x}}_{t-1}$, add a measurement-consistency gradient:

$$\mathbf{x}_{t-1} \leftarrow \hat{\mathbf{x}}_{t-1} - \zeta_t \nabla_{\mathbf{x}_t} \|\mathbf{y} - \mathbf{A}\hat{\mathbf{x}}_0(\mathbf{x}_t)\|^2,$$

where $\hat{\mathbf{x}}_0(\mathbf{x}_t)$ is the current x_0 -estimate from the denoiser and ζ_t is a step-size schedule. This is analogous to “diffusion posterior sampling” (DPS).

- (c) The per-element uncertainty map is $u_i = \sqrt{\frac{1}{S-1} \sum_{s=1}^S (\hat{x}_{0,i}^{(s)} - \bar{x}_{0,i})^2}$, i.e. the empirical standard deviation across samples for each component i . Components with high u_i are poorly constrained by measurements and should be flagged as unreliable.
- (d) Higher σ_n makes each measurement less informative, so the posterior is broader. The diffusion samples will be more diverse because the measurement-consistency gradient is weaker relative to the prior.

D3. DiffusionDet-Style Object Detection

- (a) Bounding-box coordinates (x_c, y_c, w, h) are continuous values, just like pixel intensities. The DDPM forward process only requires the data to be continuous and approximately normalizable. After standardizing box coordinates to zero mean and unit variance, the Gaussian corruption is well-defined.
- (b) Given S independent runs, each produces a set of refined boxes. For a particular object, the spread of the predicted (x_c, y_c, w, h) across samples quantifies localization uncertainty. High spread in (x_c, y_c) indicates uncertain object center; high spread in (w, h) indicates uncertain object extent.
- (c) A detection could be deemed “reliable” if: (1) the classification confidence exceeds a threshold (e.g. 0.9), and (2) the standard deviation of the bounding-box coordinates across S samples is below a task-specific tolerance (e.g. $\text{std}(x_c) < 0.05 \times \text{image width}$). Detections failing either criterion are deferred to a human operator.

D4. Multimodal Conditioning: Sensor Fusion

- (a) **Option 1 (Concatenation):** Encode $\mathbf{c}_1$ and $\mathbf{c}_2$ into embeddings and concatenate them (or tile spatially) alongside $\mathbf{x}_t$ as additional input channels to the U-Net.
Option 2 (Cross-attention): Encode each condition with a separate encoder, then inject them via cross-attention layers in the U-Net (similar to how text is injected in Stable Diffusion). This is more flexible and allows the model to attend selectively to each modality.
- (b) During training, randomly drop $\mathbf{c}_1$, $\mathbf{c}_2$, both, or neither (with some probability each). At test time, if $\mathbf{c}_2$ is unavailable, set $\mathbf{c}_2 = \emptyset$ and optionally apply guidance with respect to $\mathbf{c}_1$ only. The model has learned to operate with partial conditioning.
- (c) Generate S samples, compute per-pixel sample mean and standard deviation. Compare the sample standard deviation to the actual absolute error $|\hat{x}_i - x_{0,i}^{\text{true}}|$ across a test set. A well-calibrated model should have the empirical coverage of its prediction intervals match the nominal level (e.g. 90% of true values fall within the 5th–95th percentile band).

D5. Diffusion for Signal Inpainting

- (a) **Repaint algorithm:** At each reverse step t , after computing the denoised estimate $\mathbf{x}_{t-1}$:
 - For observed indices ($M_i = 1$): replace with a noised version of the true observed values at the appropriate noise level: $x_{t-1,i} \leftarrow \sqrt{\bar{\alpha}_{t-1}} x_{\text{obs},i} + \sqrt{1 - \bar{\alpha}_{t-1}} \epsilon_i$.
 - For missing indices ($M_i = 0$): keep the model’s prediction.This ensures observed values remain consistent while the model fills in the gaps.
- (b) A short missing block (8 samples) is heavily constrained by the surrounding observed context. The conditional distribution is narrow, so sample diversity is **small**.

- (c) A long missing block (128 samples) has far less contextual constraint. Multiple plausible completions exist, so sample diversity is **large**. This larger spread reflects genuine aleatoric ambiguity: many signals are consistent with the observed boundaries.
- (d) Classical linear interpolation produces a single deterministic fill with no uncertainty estimate and cannot capture nonlinear signal structure. Autoregressive models generate one sample at a time in a fixed direction and may accumulate errors. Diffusion-based inpainting produces multiple diverse samples, naturally handles bidirectional context (past and future of the gap), and provides nonparametric uncertainty estimates.

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